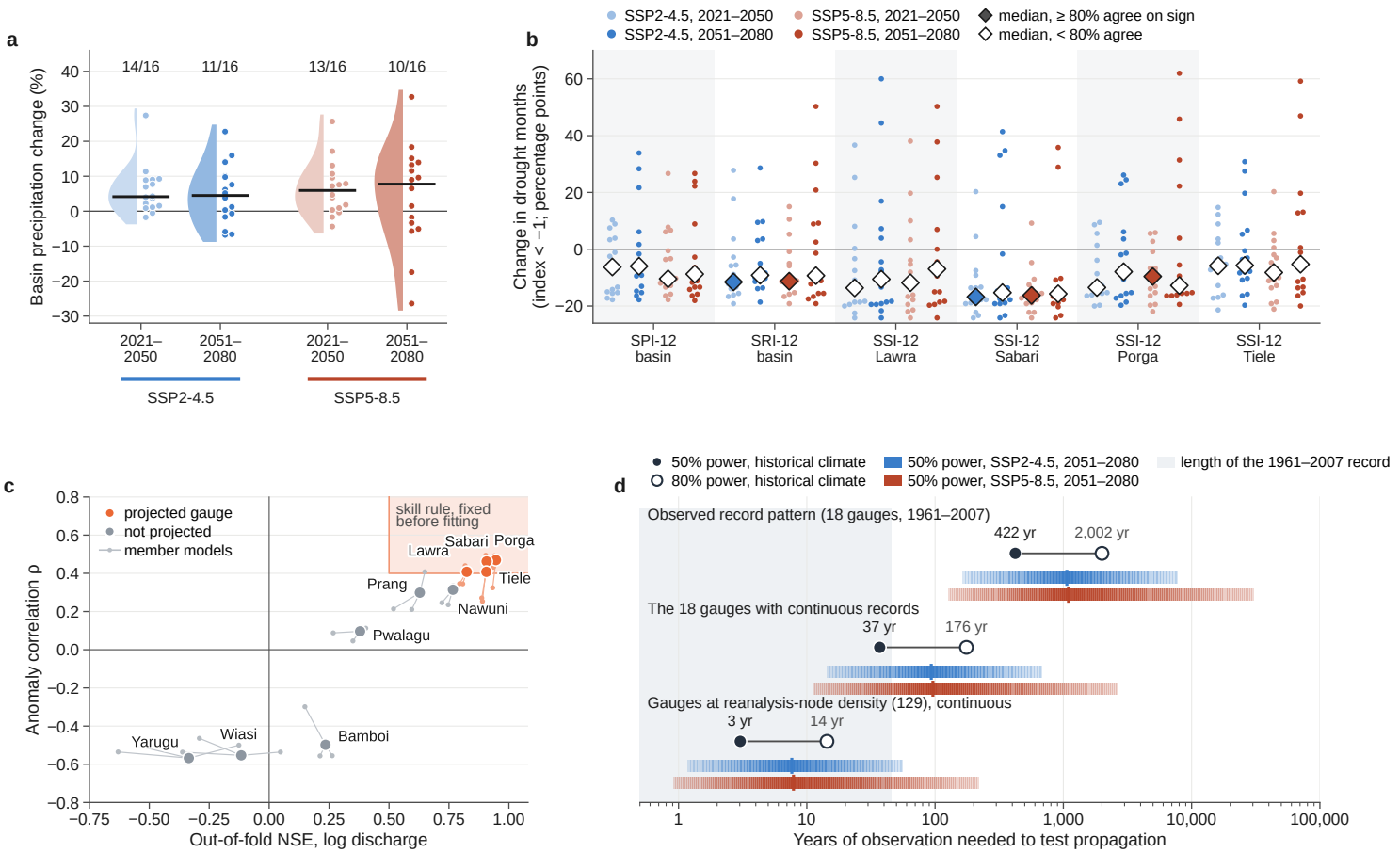

a
14/16
11/16
13/16
10/16
Basin precipitation change (%)
40
30
20
10
0
−10
−20
−30
2021–2050
2051–2080
2021–2050
2051–2080
SSP2-4.5
SSP5-8.5
b
SSP2-4.5, 2021–2050
SSP2-4.5, 2051–2080
SSP5-8.5, 2021–2050
SSP5-8.5, 2051–2080
median, ≥ 80% agree on sign
median, < 80% agree
Change in drought months
(index < −1; percentage points)
60
40
20
0
−20
SPI-12 basin
SRI-12 basin
SSI-12 Lawra
SSI-12 Sabari
SSI-12 Porga
SSI-12 Tiele
c
projected gauge
not projected
member models
skill rule, fixed before fitting
Lawra
Sabari
Porga
Tiele
Prang
Nawuni
Pwalagu
Yarugu
Wiasi
Bamboi
Anomaly correlation ρ
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.8
−0.75
−0.50
−0.25
0.00
0.25
0.50
0.75
1.00
Out-of-fold NSE, log discharge
d
50% power, historical climate
80% power, historical climate
50% power, SSP2-4.5, 2051–2080
50% power, SSP5-8.5, 2051–2080
length of the 1961–2007 record
Observed record pattern (18 gauges, 1961–2007)
422 yr
2,002 yr
The 18 gauges with continuous records
37 yr
176 yr
Gauges at reanalysis-node density (129), continuous
3 yr
14 yr
1
10
100
1,000
10,000
100,000
Years of observation needed to test propagation